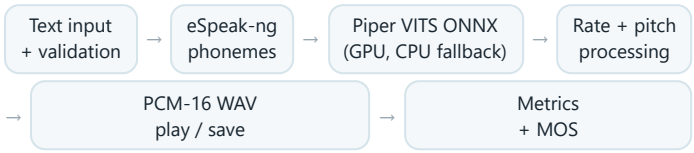
Assignment 2 — Personalized Text-to-Speech System

Speech Processing (S2-25 AIMLCZG520) | Project 4 (recommended) | Group 16 | Hamza Aziz (2024AC05133)

OBJECTIVE

Build an application that converts user text into natural-sounding speech using genuine male/female neural voices, independent rate and pitch controls, in-app playback and WAV export. Inference prefers GPU (CUDA) execution and falls back to CPU automatically, so the same code runs unchanged on a Windows workstation and on the GPU-less Rocky Linux 9.5 BITS Virtual Lab. Evaluate response time, objective signal diagnostics, and genuine listener MOS.

SYSTEM ARCHITECTURE / WORKFLOW



TOOLS, LIBRARIES, AND TECHNOLOGIES USED

Python 3.14.6; Piper TTS 1.6.0; Streamlit 1.61.1; librosa 1.0.0; SoundFile 0.14.0; NumPy 2.5.2; Matplotlib 3.11.1; ONNX Runtime GPU 1.28.0 (CUDA execution provider preferred, CPU provider as automatic fallback). Voices: en_US-hfc_female-medium and en_US-hfc_male-medium (22,050 Hz; CC BY-NC-SA 4.0).

IMPLEMENTATION DETAILS AND KEY FEATURES

1. Input normalization, empty/length/range validation, and actionable missing-model errors; character limit is UI-configurable (default 5,000).
2. Two authentic single-speaker models; gender is not simulated by pitch.
3. Rate multiplier maps to length_scale = 1/rate; pitch is transparently post-processed in semitones because Piper has no native pitch field.
4. GPU (CUDA) preferred with an automatic, self-tested fallback to CPU. Sentence audio chunks stream in with a live progress bar, are concatenated, clipped safely, encoded as PCM-16 WAV, played, downloaded, and saved with JSON metadata.
5. Innovation: cached ONNX loading, waveform/log-mel dashboard, RTF and quality diagnostics, streaming synthesis progress, listener MOS capture, and reproducible negative tests.

CHALLENGES FACED

Voice models total about 126 MB and introduce a cold start; model loading is cached and measured separately. The first pitch-shift kernel has a one-time compile cost, so explicit warm-up prevents biased steady-state timing. Enabling GPU execution on Windows required exact-matched cuDNN/cuBLAS wheel versions (onnxruntime-gpu 1.28 + cuDNN 9.8.0.87 + cuBLAS 12.8.4.1); a mismatch fails silently mid-inference, so voice loading now runs a real warm-up synthesis and falls back to CPU on any failure. A PDF cannot carry playable audio, so four WAV files accompany the notebook. Extreme rate/pitch settings can reduce naturalness.

RESULTS AND OBSERVATIONS

Measured on the BITS Virtual Lab (Rocky Linux 9.5, CPU fallback — no GPU present). Four configured samples: mean synthesis response = 0.749 s; mean RTF = 0.163; maximum clipping = 0.00096%; all outputs = 22,050 Hz WAV.

Seven repeated warm runs: median latency = 0.471 s; p95 = 0.816 s; median RTF = 0.101. RTF below 1 means generation is faster than playback, even on CPU.

Cold model loads: female = 2.65 s; male = 2.54 s.

Moderate rate and pitch changes produced distinct durations/spectral profiles without material clipping. Four invalid-input tests and all 15 notebook code cells passed with zero errors, confirming the GPU-preferred code runs correctly on its CPU fallback path in the actual Virtual Lab environment.

EVALUATION STATUS AND RESPONSIBLE USE

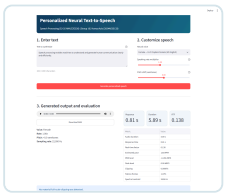
MOS: 5 genuine listener ratings collected via the Streamlit form on the BITS Virtual Lab, covering both voices and three rate/pitch configurations (1.00×/+0 st, 1.05×/-0.5 st, 1.40×/+4 st): mean naturalness = 4.0/5 (std 1.23), clarity = 3.8/5 (std 1.30), overall = 4.0/5 (std 1.00), n=5. No rating was fabricated; the lowest score (2/5) was for an extreme 1.40× rate, consistent with the rate/pitch caveat below. Objective RMS, peak, clipping, silence, centroid, and zero-crossing values are diagnostics, not substitutes for perceptual judgment.

Use: Outputs are labelled synthetic; the academic models are not intended for impersonation or commercial deployment.

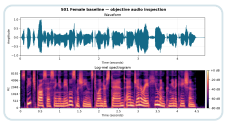
SOURCE AND REPRODUCIBILITY

Requirements: official Problem-Statement.pdf, pages 1 and 4. The executed notebook (`notebooks/`), modular app/core source, pinned Pipfile/requirements, model downloader, sample CSV, benchmark CSV/JSON, figures, audio, and run instructions are included. GPU execution is preferred; CPU execution requires no CUDA and works unchanged on Rocky Linux 9.5, and neither path needs FFmpeg.

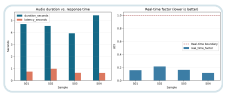
SAMPLE OUTPUT SCREENSHOTS



A. Streamlit generated state



B. Waveform + log-mel



C. Duration/response/RTF

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